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CV for Dr Clare Rainey
Valid July 2026

Personal Details

Name	Dr Clare Rainey, PhD, PgDip, FHEA, PgCHEP, BSc (Hons)
Business Address	Discipline of Radiography, School of Medicine, University College Cork, Cork, Ireland
Telephone	07834855438
E-mail	clare.rainey@ucc.ie
h-index	11 (Clare Rainey - Google Scholar https://scholar.google.com/citations?hl=en&user=hvHG8mEAAAAJ&view_op=list_works&sortby=pubdate) 554 citations since 2021

Dr Clare Rainey is an experienced academic with a PhD in AI for radiographic image interpretation. She has experience in a number of statistical analysis methodologies for both qualitative and quantitative studies. Clare has extensive expertise in research supervision, curriculum development, and academic assessment. She has served as an External Examiner for MSc programs and has a strong track record in National and International scholarly contributions. Her leadership roles in AI research and radiographer education demonstrate her ability to uphold academic standards and provide rigorous quality assurance in academia and clinical practice.

Experience and Research Interests

- Artificial Intelligence in health and medicine
- AI in education
- Frontier AI models and impact on users
- Human computer integration
- Responsible AI
- Student engagement
- Learning innovation
- Professional leadership
- Research supervision
- Academic writing

Education

2023	PhD, Ulster University
2019	PgC Higher Education Practice, Ulster University
2015	PgD Advancing Practice in an AHP Specialism, University of Ulster
2013	PgC Medical Imaging (Computed Tomography), University of Bradford
2005	BSc (Hons) Diagnostic Radiography and Imaging, University of Ulster

Career History

2025 – present	Founding member of the European Federation of Radiographer Societies (EFRS) AI and Informatics Working Group
2025 – present	Founder and Chair, CHERISH-AI (Collaborative hub for emerging research and innovation in smart health and AI in Ireland)
2025 – present	Chair, Society and College of Radiographers AI Advisory Board
2025 – present	Associate editor *Journal of Medical Imaging and Radiation Sciences*
2025 – present	Lecturer in Medical Imaging and Radiation Sciences, UCC

2021 – 2025	Course Director/Programme lead, Diagnostic Radiography and Imaging, Ulster University
2017 – 2022	Lecturer in Diagnostic Radiography and Imaging, Ulster University
2016 (05/12) - 2017	Teaching Fellow in Diagnostic Radiography and Imaging, Ulster University
2012 – 2016 (04/12)	Part-time Lecturer and Marker (general radiography), Ulster University
2011 – 2015	Part-time Lecturer and Marker (Pg. Computed Tomography), University of Bradford
2008 – 2016	Band 6 CT and General Radiographer, Causeway Hospital
2007 – 2008	Band 5 General Radiographer, Causeway Hospital
2005 – 2007	Link – Grade Orthopaedic Radiographer, Musgrave Park Hospital

Recent Research Grants

Enterprise Ireland Horizon Europe Co-ordinator Proposal Preparation Support Scheme for submission of MSC DN in Nov **2026** entitled TRUST-Health AI – Training Researchers for User-Safe and Trustworthy Health AI (€12,192)

'An observational study to investigate the impact of AI feedback on image comment, decision switching and trust perception in student and experienced radiographers (£10,000) *PI C Rainey (PI),* Dr S McFadden, Dr J McConnell, Prof. C Hughes, Dr R Bond. CoRIPS (College of Radiographers Industry Partnership Scheme) (**2020-2023).**

'An experimental study to investigate the impact of different forms of AI feedback on automation bias and decision switching on diagnostic radiographers' *PI Dr Clare Rainey*, Dr S McFadden, Dr J McConnell, Prof. C Hughes, Dr R Bond (£10,000) CoRIPS (College of Radiographers Industry Partnership Scheme) (**2022-2023**)

'Radiographer EducAtion and Learning in AI- (REAL AI)' *PI Dr S McFadden*, Dr J McConnell, Prof. C Hughes, Dr R Bond, Dr C Rainey, Dr L McLaughlin. (£10,000) CoRIPS (College of Radiographers Industry Partnership Scheme) (**2022-2024**)

An evaluation of a checklist in musculoskeletal radiographic image interpretation when using Artificial Intelligence (AI) *PI Dr L McLaughlin*, Dr S McFadden, Dr C Rainey, Dr J McConnell, Prof. C Hughes, Dr R Bond (£10,000) CoRIPS (College of Radiographers Industry Partnership Scheme) (**2022-2024**)

An analysis of different user interfaces and designs for clinically relevant decision support when using AI: what do imaging professionals prefer? Avneet Gill, Dr Clare Rainey, Dr Laura McLaughlin, Dr Ciara Hughes, Dr Raymond Bond, Prof Jonathan McConnell, Dr Sonyia McFadden (£5000) UKIO Pitch your Project Competition, UKIO **2024**

Publications

Recent relevant publications (full list via Google Scholar)

Papers and book contribution

Rainey C, McLaughlin L, England A, Malamateniou C, McFadden SL, Woznitza N. Generative AI and large language models in radiography education: Possibilities, obstacles, and expectations for academic staff. Journal of Medical Imaging and Radiation Sciences. 2026 Jul 1;57(4).

Book chapter: Rainey C. and Gill, A. Balancing trust and reliance: Understanding the human-AI interaction to ensure responsible use of innovation and advanced technologies in radiography. In: Artificial Intelligence and Data Analytics in Medical Imaging 2026 May 19 (pp. 81-93). CRC Press.

Rainey C, McFadden S, Gill A. Radiology artificial intelligence for prioritized imaging and diagnosis of lung cancer: qualitative interview analysis of stakeholder perspectives in Northern Ireland. Frontiers in Medicine. 2026 Apr 2;13:1759041.

England A, Rainey C, McLaughlin L, McEntee M, Societies R. Simulation in Diagnostic Radiography Training: A Replacement for Clinical Placements?. Radiology. 2026 Apr.

BOOK (lead editor) : Rainey C, McFadden S, McConnell J, editors. Digital Healthcare and Artificial Intelligence: A Primer for Clinicians and Students. CRC Press; 2026 Feb 2.

Book chapter: McNally C, Rainey C, Szczepura K. AI in Projectional Radiography. InArtificial Intelligence for Radiographers: Basic Principles, Clinical Applications and Implementation Considerations 2026 Jan 15 (pp. 85-118). Cham: Springer Nature Switzerland.

Busch F, Hoffmann L, Xu L, et al. (2025) Multinational Attitudes Toward AI in Health Care and Diagnostics Among Hospital Patients. *JAMA Netw Open.* 8(6):e2514452. doi:10.1001/jamanetworkopen.2025.14452

McLaughlin, L., McFadden, S.L., Villikudathil, A.T., McConnell, J., Hughes, C., Bond, R. and Rainey, C. (2025), An evaluation of a checklist in musculoskeletal radiographic image interpretation when using artificial intelligence. J Med Radiat Sci, 72: 209-216. https://doi.org/10.1002/jmrs.850

Gill, A., Rainey, C., McLaughlin, L., Hughes, C., Bond, R., McConnell, J. and McFadden, S., 2025. Artificial Intelligence user interface preferences in radiology: A scoping review. *Journal of medical imaging and radiation sciences*, *56*(3), p.101866.

Creeden, A., McFadden, S., Rainey, C., Campbell, S., Ather, S., Hajilou, A., Bond, R., McAllister, P. and Woznitza, N., 2025. Accuracy of interpretation of nasogastric tube position on chest radiographs by diagnostic radiographers: A multi-case, multi-reader study. *Radiography*, *31*(1), pp.83-88.

Rainey, C., Bond, R., McConnell, J., Gill, A., Hughes, C., Kumar, D. and McFadden, S., 2025. The impact of AI feedback on the accuracy of diagnosis, decision switching and trust in Radiography. *PLoS One*, *20*(5), p.e0322051.

Rainey C, Bond R, McConnell J, Hughes C, Kumar D, McFadden S (2024) Reporting radiographers' interaction with Artificial Intelligence—How do different forms of AI feedback impact trust and decision switching? PLOS Digit Health 3(8). https://doi.org/10.1371/journal.pdig.0000560

Rainey, C., 2024. Artificial intelligence and radiographer preliminary image evaluation: What might the future hold for radiographers providing x-ray interpretation in the acute setting?. *Journal of Medical Radiation Sciences*, *71*(4), pp.495-498

Rainey, C., Bond, R., McConnell, J., Hughes, C., Kumar, D. and McFadden, S., 2024. Reporting radiographers' interaction with Artificial Intelligence—How do different forms of AI feedback impact trust and decision switching?. *PLOS Digital Health*, *3*(8), p.e0000560.

Walsh G, Stogiannos N, van de Venter R, Rainey C, Tam W, McFadden S, McNulty JP, Mekis N, Lewis S, O'Regan T, Kumar A, Huisman M, Bisdas S, Kotter E, Pinto Dos Santos D, Sá Dos Reis C, van Ooijen P, Brady AP, Malamateniou C. (2023) Responsible AI practice and AI education are central to AI implementation: a rapid review for all medical imaging professionals in Europe. BJR Open. https://doi.org/10.1259/bjro.20230033

Rainey, C., Villikudathil, A., McConnell, J., Hughes, C., Bond, RR. & McFadden, S. (2023) An experimental machine learning study investigating the decision-making process of students and qualified radiographers when interpreting radiographic images. PLoS Digital Health. 2 (10) https://doi.org/10.1371/journal.pdig.0000229

O'Neill, A., Hughes, C., McClure, P., Rainey, C., McLaughlin, L. & McFadden, S. (2023) Patient engagement with radiation therapists: Patient perspectives, challenges, and opportunities. A Systematic review. Radiography. 29. https://doi.org/10.1016/j.radi.2023.02.022

Rainey, C., O'Regan, T., Matthew, J., Skelton, E., Woznitza, N., Chu, K-Y., Goodman, S., McConnell, J., Hughes, C., Bond, RR., Malamateniou, C. & McFadden, S. (2022) UK reporting radiographers' perceptions of AI in radiographic image interpretation – Current perspectives and future developments. Radiography. 28 (4) https://doi.org/10.1016/j.radi.2022.06.006 ***** Award for Top 10 most downloaded articles in 2023 *****

Rainey, C., O'Regan, T., Matthew, J., Skelton, E., Woznitza, N., Chu, K-Y., Goodman, S., McConnell, J., Hughes, C., Bond, RR., Malamateniou, C. & McFadden, S. (2022) An insight into the current perceptions of UK Radiographers on the future impact of AI on the profession: a cross-sectional survey. The Journal of Medical Imaging and Radiation Sciences. 53 (3) https://doi.org/10.1016/j.jmir.2022.05.010 ***** Editors' choice – Top 5 articles of the year *****

Tekin, H. O., Rainey, C., AlMisned, G., Issa, S. A. M., Akkus, B. & Zakaly, H. M. H. (2022) Heavy metal oxide added glassy portable containers for nuclear waste management applications: In comparison with reinforced concrete containers. Radiation Physics and Chemistry. 201, 110449. https://doi.org/10.1016/j.radphyschem.2022.110449

Rainey, C., McConnell, J., Hughes, C., Bond, RR., & McFadden, S. (2021). Artificial Intelligence for diagnosis of fractures on plain radiographs: a scoping review of current literature. Intelligence-Based Medicine, 5. https://doi.org/10.1016/j.ibmed.2021.100033

Rainey, C., O'Regan, T., Matthew, J., Skelton, E., Woznitza, N., Chu, K-Y, Goodman, S., McConnell, J., Hughes, C., Bond, R., McFadden, S., Malamateniou, C. (2021) Beauty is in the AI of the beholder: are we ready for the clinical integration of artificial intelligence in Radiography? An exploratory analysis of perceived AI knowledge, skills, confidence and education perspectives of UK radiographers. Frontiers in Digital Health https://doi.org/10.3389/fdgth.2021.739327

Oral presentations

Rainey C (2025) Human and AI Interaction. EuSoMII 2025, Crete (invited)

Rainey, C., Bond, RR., Hughes, C., McConnell, J. McFadden, S., (2023) The impact of AI feedback on the accuracy of diagnosis, decision switching and trust in Radiography. ECR 2023, Vienna.

McLaughlin, L., Rainey, C., Hughes, C., McConnell, J., Bond, RR., Villikudathil, A., & McFadden, S. (2023). An evaluation of a checklist in Musculoskeletal (MSK) radiographic image interpretation when using Artificial Intelligence (AI). 85, P103. Abstract from UK Imaging & Oncology Congress 2023, Liverpool, United Kingdom.

Doherty, G., Mc Laughlin, L., Rainey, C., Hughes, C., Bond, RR., McConnell, J., & McFadden, S. (2023). Radiographer/radiologist education and learning in artificial intelligence (REAL-AI). 28. Abstract from UK Imaging & Oncology Congress 2023, Liverpool, United Kingdom.

Rainey, C., O'Regan, T., Matthew, J., Skelton, E., Woznitza, N., Chu, K-Y, Goodman, S., McConnell, J., Hughes, C., Bond, R., McFadden, S., Malamateniou, C., (2021) Beauty is in the AI of the beholder: Are we ready for the clinical integration of artificial intelligence in Radiography? An analysis of AI knowledge, skills, confidence and education perspectives of UK radiographers.' ISRRT 20-22 August, Dublin (online)

Rainey, C., McFadden, S., Bond, R. (2021) Digital Divides in healthcare education innovations. Medical Education Informatics Conference. 12-15 July. Nottingham, UK (online) ** ***Bronze award for 'excellent presentation.' *****

Rainey, C., Williamson, K. (2020) Developing clinical competencies in post pandemic higher education. Simulation in Radiography Teaching Conference (international). 24 June 2020. Online

Poster presentations

Rainey C, McLaughlin L, McEntee M, England A, Gill A, McConnell J, mc Fadden S (2025) Clinicians' perceptions of Large Language Model (LLM) outputs for simplification of radiology reports of the appendicular skeleton. EuSoMII, Crete, 2025

Keys A, Duffy O, Rogers R, Rainey C (2025) Artificial Intelligence in Videofluroscopy and Fibreoptic Endoscopic Evaluation of Swallowing Analysis: A Scoping review. EuSoMII, Crete, 2025

Gill A, Hughes C, McConnell J, Rainey C, McLaughlin L, Bond R, Mc Fadden S (2025) Advancing AI integration within radiology through multidisciplinary collaboration: the impact of an AI Study Conference. EuSoMII, Crete, 2025

Gill A, Hughes C, McConnell J, Rainey C, McLaughlin L, Bond R, Mc Fadden S (2025) Think Aloud Study for Knowledge (TASK) about AI user interface perceptions. EuSoMII, Crete, 2025

Doherty G, Hughes C, McConnell J, Rainey C, McAlister P, McLaughlin L, Bond R, Mc Fadden S (2025) AI education intervention for medical imaging professionals: development, delivery and outcomes. EuSoMII, Crete, 2025

Rainey C, Gill A, McFadden S (2025) Radiology AI for Prioritised Imaging and Diagnosis of Lung Cancer (RAPID-LC): the stakeholders perspective. EuSoMII, Crete, 2025

Rainey, C., Hughes, C., Bond, RR., McConnell, J. & McFadden, S. (2024) The impact of forms of AI feedback and image quality on reporting radiographers trust and decision switching when interpreting plain radiographic images of the appendicular skeleton. ECR 2024, Vienna. https://dx.doi.org/10.26044/ecr2024/C-23132

Rainey, C., Matthew, J., Skelton, E., Woznitza, N., Chu, K_Y., Goodman, S., McConnell, J., Hughes, C., Bond, R., O'Regan, T., Malamateniou, C., McFadden, S. (2022) 'UK radiographers' and radiotherapists' perceptions and expectations of AI in radiology – current status and future developments' European Congress of Radiology, Vienna

Rainey, C. *(presenter – 4 min oral presentation of poster),* McConnell, J., Bond, R., Hughes, C., McFadden, S. (2021) Human vs Machine: A comparison of the differences of AI and human vision interpretation methods. ISRRT, 20-22 August, Dublin (online).

Shand, C., Rainey, C., (2019) Nearpod® in Diagnostic Radiography Education: Student engagement and feedback. UKIO, 10-12 June 2019, Liverpool

Other

Doherty, G., Bond, RR., Hughes, C., Rainey, C., Mc Laughlin, L., McAllister, P., Mc Fadden, S., & McConnell, J. (2023, Oct 27). An introduction to artificial intelligence for healthcare professionals. Advance online publication (NHS E-learning for health platform). https://portal.e-lfh.org.uk/Component/Details/828621

Doherty, G. (Author), Bond, RR. (Author), Mc Laughlin, L. (Author), Hughes, C. (Author), Rainey, C. (Author), McConnell, J. (Author), & McFadden, S. (Author). (2023). General Introduction to AI for Radiographers. Digital or Visual Products, The Society of Radiographers. Advance online publication. https://www.sor.org/learning-advice/learning/on-demand/webinars/general-introduction-to-ai-for-radiographers

Rainey, C. (2022). Leading by living – A personal journey towards values and goals-based leadership of self and others. The Journal of Medical Imaging and Radiation Sciences, 53(4), S32-S34. https://doi.org/10.1016/j.jmir.2022.08.009

Conference organisation / chair

Conference Organisation / Chair: Rainey, C., McFadden, S., Bond, R. (2021) Educating for a Digital Future: what do healthcare professionals need to know about AI and digital health? 14 April 2021. Online. Available at: https://youtube.com/playlist?list=PL2-fB0_yayJ-mlcHvV-a5d5hsGXfoDnEM

Conference organisation/Chair: Gill, A., Rainey, C., McConnell, J., McFadden, S., (2024) NI Healthcare AI study day: Ulster University. Planning stages – due to run on 25th and 26th November 2024.

Auditor/Quality assurance

Auditor for the American Board of AI in Medicine (ABAIM) Advanced programme (DATE)

Professional affiliations/positions

Member of AIMed 2020 – 2024
Certification from the American Board of AI in Medicine (ABAIM) (2021)
Member of the SCoR AI and Emerging Technologies Advisory Group 2020 – present
Appeals Panel member Health and Care Professions Council (HCPC) (2021-2025)
Member of the Association of Learning Technologists (ALT) 2017 – present
Fellow of the Higher Education Academy (HEA) (2019-current)
B.I.R. Conference Abstract Editor (member of Abstract Review Committee) (2021 - present)
Nearpod® Certified Educator 2017 – present
Associate Editor – Journal of Medical Imaging and Radiation Sciences

Other

External Examiner MSc Glasgow Caledonian University 2023 - present
Chair and founder of the SUCCESS UU group – development of a Health Profession specific student charter 2023 – 2025
Founding member of the Simulated Learning Group (Ulster University) - 2024
Founding member of the Service User Involvement Group (Ulster University) - 2024
Faculty member – DEXA committee – 2020-2024
Faculty member – Health and Safety – 2020-2024
UU Faculty member – Teaching and Learning committee – 2020-2024
UU Health and Safety co-ordinator (School of Health Sciences) 2020 - 2024
UU Radiation Protection Supervisor (RPS) 2020 – AY 2023/4
UU Member of the School of Health Sciences Transition Board (responsible for strategic planning of the move of the School to a new campus – 2022)
Ulster University, Faculty of Life and Health Sciences Revalidation panel - present
Sentinus® Young Scientist/Young Research and Development Judge (2019 and 2023)